

* Home Assignment :- 2 *

Q1] Which subsets are subspaces of vector space $\mathbb{R}^2$ Plane $V = \mathbb{R}^2 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x, y \in \mathbb{R} \right\}$.

→ 4)

$$W = \left\{ \begin{bmatrix} x \\ 3x+2 \end{bmatrix} \mid x \in \mathbb{R} \right\}$$

$$\therefore \begin{bmatrix} x \\ 3x+2 \end{bmatrix} = x \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\text{Set } x = 0 \Rightarrow \text{vector} = [0, 2] \neq [0, 0]$$

$\therefore$ Not a subspace.

$\therefore W$ is not subspace because it doesn't contain zero vector.

Q2] Which subsets are subspaces of $V = P_3(t) = \{ a + bt + ct^2 + dt^3 \mid a, b, c, d \in \mathbb{R} \}$

→ 2)

$$W = \{ a \mid a \in \mathbb{R} \}, \text{ it is a subspace of } P_3(t) \\ \text{--- (polynomial of degree } \leq 3 \text{)}$$

It contains 0 polynomial, ($a=0$)

closed under addition, $a_1 + a_2 \in \mathbb{R}$

closed under ~~add~~ scalar multiplication.

$\therefore$ W is a subspace of $P_3(t)$ & is space of all constant polynomials.

Dimension $\Rightarrow 1$ ($\because$ it's spanned by basis $\{1\}$).

Q3] Examine if vectors forms a basis & if so find dimension.

→ 4) $W = \{t, t^2, t^3\}$ of $P_3(t) = \{a+bt+ct^2+dt^3 \mid a, b, c, d \in \mathbb{R}\}$

Assume that they are linearly independent,
 $c_1 t + c_2 t^2 + c_3 t^3 = 0$

for all $t \Rightarrow$ Each coefficient = 0

$\therefore \cancel{c_1} = 0, c_1 = 0, c_2 = c_3 = 0$

$\Rightarrow$ Linearly independent.

$P_3(t)$ has dimension 4 $\rightarrow$ needs 4 basis elements.
 These are only 3 vectors.

Thus, they form linearly ind. set.
 but not a basis of $P_3(t)$.

& They span a subspace of dimension 3.

Q4] Find basis & dimension of subspaces -
 Row(A), Col(A), & Null(A)

→ 3)

$$A = \begin{bmatrix} 1 & 2 & -1 & 0 & 3 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Row space:

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 & 3 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$\therefore$ Rows are linearly independent.

Row space basis = all 3 rows

Dimension = 3.

column space:

Pivot elements in RREF: columns $\Rightarrow 1, 3 \& 4$.

$$C_1 = [1, 0, 0]$$

$$C_3 = [-1, 1, 1]$$

$$C_4 = [0, 0, 1]$$

$\therefore$ col space basis = $\{C_1, C_3, C_4\}$

Dimension = 3.

Null space:

$$A \cdot x = 0$$

From RREF,

$$x_1 + 2x_2 - x_3 + 0x_4 + 3x_5 = 0$$

$$x_3 + x_5 = 0$$

$$\Rightarrow x_3 = -x_5$$

$$x_4 = 0$$

Free variables $\Rightarrow x_2 \& x_5$

General solⁿ $\Rightarrow$

$$x = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

Null space $\Rightarrow$ two vectors.

Dimension $\Rightarrow 2$.